

Q5a) Illustrate with an example the use of Unification Algorithm to prove the concept of Resolution.

Unification is the process of making two logical expressions identical by finding suitable substitutions for variables.

The substitutions obtained are called: Unifier, Most General Unifier (MGU)

Need of Unification - In First Order Logic, predicates may contain variables.

To apply resolution: Predicates must match exactly.

Unification helps in matching predicates.

Resolution in FOL

Resolution is an inference rule used to derive conclusions by eliminating complementary literals.

Resolution Rule: From: $P \vee Q$ and $\neg Q \vee R$

Infer: $P \vee R$

Unification Algorithm - Steps:

1. Compare predicates.
2. Match variables with constants.
3. Find substitutions.
4. Apply substitutions.
5. Resolve matching literals.

Example - Statements -

1. $\forall x[Human(x) \rightarrow Mortal(x)]$
2. $Human(Socrates)$

Goal: $Mortal(Socrates)$

Step 1: Convert into Clause Form

Statement 1 - Remove implication: $\neg Human(x) \vee Mortal(x)$

Statement 2 - $Human(Socrates)$

Negated Goal - $\neg Mortal(Socrates)$

Step 2: Apply Unification

Compare: $Mortal(x)$ and $Mortal(Socrates)$

Substitution: $x = Socrates$

This is the unifier.

Step 3: Resolution

Resolve: $\neg \text{Human}(\text{Socrates}) \vee \text{Mortal}(\text{Socrates})$ with $\neg \text{Mortal}(\text{Socrates})$

Result: $\neg \text{Human}(\text{Socrates})$

Now resolve with: $\text{Human}(\text{Socrates})$

Result: $\square$ (Empty clause)

Hence proved: $\text{Mortal}(\text{Socrates})$

Advantages of Unification - Simplifies matching process, Enables automated reasoning, Important in theorem proving, Used in Prolog and AI systems

Applications - Expert systems, Logic programming, NLP, Automated theorem proving

Q5b) Define and explain Forward Chaining with example. Analyze differences between Forward and Backward Chaining.

Forward Chaining is a data-driven inference technique in which reasoning starts from known facts and applies rules repeatedly until goal is reached.

Working of Forward Chaining

1. Start with known facts.
2. Apply inference rules.
3. Generate new facts.
4. Continue until goal is achieved.

Structure of Rule - IF condition
THEN conclusion

Example -

Facts: Rain

Rule: $\text{Rain} \rightarrow \text{WetRoad}$

Process - Since Rain is true: Infer: WetRoad

Forward Chaining Algorithm

1. Add known facts to KB
2. Match rules whose conditions are true
3. Fire rule

4. Add new facts
5. Repeat until goal reached

Advantages of Forward Chaining - Simple reasoning method, Useful when many facts are available, Automatically derives conclusions

Applications - Expert systems, Medical diagnosis, Rule-based AI systems

Difference between Forward and Backward Chaining

Forward Chaining	Backward Chaining
Data-driven	Goal-driven
Starts from facts	Starts from goal
Generates all conclusions	Searches only needed facts
Suitable when many facts exist	Suitable for specific query
Bottom-up approach	Top-down approach
Used in monitoring systems	Used in diagnosis systems

Forward chaining derives conclusions from available data. Backward chaining starts with goal and searches supporting facts. Both are important inference techniques in AI and expert systems.

Q6a) Describe Ontological Engineering with respect to Categories, Objects and Models.

Ontological Engineering is the process of designing and organizing knowledge in a structured form so that AI systems can understand and reason about the world.

It involves: Defining concepts, Relationships, Objects, Properties, used in a domain.

Ontology - An ontology is a formal representation of knowledge containing: Objects, Categories, Relations, Properties, Rules

Need of Ontological Engineering - Knowledge sharing, Reusability of knowledge, Intelligent reasoning, Semantic understanding, Standardized representation

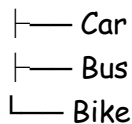
Components of Ontological Engineering

1. **Categories** - Categories are groups or classes of objects having similar properties.

Also called: Classes, Concepts

Example -

Vehicle



Here: Vehicle is category. Car, Bus, Bike are subcategories

Characteristics of Categories - Represent general concepts, Organize knowledge hierarchically, Support inheritance, Simplify reasoning

Types of Categories -

1. Natural categories - Example: Birds, Animals
2. Artificial categories - Example: Electronic Devices

2. Objects - Objects are specific individual entities belonging to categories.

Example - Car(Toyota) Car(Honda)

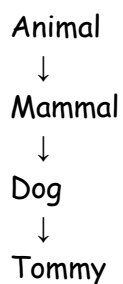
Here: Toyota and Honda are objects. Toyota and Honda are objects of category Car.

Properties of Objects - Have Identity, Possess Attributes, Participate in Relationships

3. Models - Models represent structured description of real-world entities and relationships.

They help AI systems: Interpret knowledge, Perform reasoning, Predict outcomes

Example of Ontology Model -



Tommy inherits properties of: Dog, Mammal, Animal

Types of Models

1. Semantic Networks
2. Frames
3. Logic-based Models
4. Object-based Models

Applications of Ontological Engineering - Semantic Web, Expert Systems, NLP, Medical Diagnosis, Search Engines

Advantages - Reusable knowledge, Better reasoning, Improved interoperability, Structured knowledge representation

Q6b) Define First Order Logic (FOL). Explain FOL inference for the following quantifiers/rules.

First Order Logic (FOL) is a knowledge representation technique used in AI to represent: Objects, Relationships, Properties, Quantifiers

It extends propositional logic by adding: Variables, Predicates, Quantifiers

Example - $Human(Ram)$
 $\forall x (Human(x) \rightarrow Mortal(x))$

Components of FOL - Constants, Variables, Predicates, Functions, Quantifiers

Inference Rules in FOL - Inference rules are used to derive new facts from existing facts.

i) Universal Generalization (UG)

If a statement is true for an arbitrary object, then it can be generalized for all objects.

Rule- If: $P(x)$ is true for arbitrary x , then: $\forall x P(x)$

Example - If: "Ram is mortal" $Mortal(Ram)$ then generalized as: $\forall x Mortal(x)$
(under valid conditions)

ii) Universal Instantiation (UI)

Universal Instantiation replaces universally quantified variable with a specific constant.

Rule - From: $\forall x P(x)$ infer: $P(a)$

Example - $\forall x (Human(x) \rightarrow Mortal(x))$

Infer: $Human(Ram) \rightarrow Mortal(Ram)$

iii) Existential Instantiation (EI)

Existential Instantiation replaces existential variable with a new constant.

Rule - From: $\exists x P(x)$ infer: $P(a)$ where a is a new constant.

Example - $\exists x Student(x)$

Infer: $Student(Rahul)$

iv) Existential Introduction (Existential Generalization)

If a property holds for a specific object, then we can conclude that such an object exists.

Rule - From: $P(a)$ infer: $\exists x P(x)$

Example - $Student(Ram)$ Infer: $\exists x Student(x)$

Applications of FOL - Knowledge representation, Automated reasoning, Expert systems, NLP, Intelligent agents

Q5a) Prove that Universal Instantiation is sound and that Existential Instantiation produces an inferentially equivalent knowledge base.

1. Universal Instantiation (UI) allows replacing a universally quantified variable with a specific constant.

Rule -

From: $\forall x P(x)$

infer: $P(a)$

where a is any constant.

Example - $\forall x (Human(x) \rightarrow Mortal(x))$

By UI: $Human(Ram) \rightarrow Mortal(Ram)$

Soundness of Universal Instantiation

An inference rule is sound if: Whenever premises are true, Conclusion is also true.

Proof of Soundness - Suppose: $\forall x P(x)$ is true.

Meaning: $P(x)$ is true for every object in domain.

Therefore: $P(a)$ must also be true for any particular object a .

Hence: $\forall x P(x) \models P(a)$

Therefore Universal Instantiation is sound.

Explanation - If statement says: "All birds fly" then: "Parrot flies" must also be true because parrot belongs to all birds.

2. Existential Instantiation (EI) replaces an existentially quantified variable with a new constant symbol.

Rule -

From: $\exists x P(x)$

infer: $P(c)$

where c is a new constant not appearing elsewhere.

Example - $\exists x \text{ Student}(x)$

Infer: $\text{Student}(\text{Rahul})$ where Rahul is a newly introduced symbol.

Inferentially Equivalent Knowledge Base - Existential Instantiation preserves logical meaning of knowledge base.

Meaning: Any conclusion derived from original KB can also be derived from instantiated KB.

Proof -

Suppose: $\exists x P(x)$

means: "There exists at least one object satisfying P ."

When we introduce a fresh constant c : $P(c)$

we simply name that unknown object.

No information is lost.

Thus: Original KB and instantiated KB produce same logical conclusions.

Hence EI produces inferentially equivalent KB.

Important Condition - The constant introduced must be: New, Unique

Otherwise incorrect conclusions may occur.

Applications - Automated theorem proving, Resolution in FOL, Logic programming, Expert systems

Q6a) What do you understand by Resolution? Describe the procedure to convert into CNF. Find the clause form of the expression: $(\neg P \vee Q) \rightarrow R$

Resolution is an inference rule used in Artificial Intelligence and automated theorem proving to derive conclusions from logical statements.

It works by:

1. Converting statements into clause form (CNF)
2. Resolving complementary literals
3. Producing new clauses until conclusion is proved

Resolution Rule - Given: $P \vee Q$ and $\neg Q \vee R$

Resolve on Q : $P \vee R$

Applications of Resolution - Automated theorem proving, Logic programming, Expert systems, AI reasoning systems

Procedure to Convert into CNF (Conjunctive Normal Form) - CNF is a conjunction of disjunctions of literals.

Steps for Conversion into CNF

Step 1: Eliminate Implication

Use: $A \rightarrow B \equiv \neg A \vee B$

Step 2: Move NOT inward

Apply DeMorgan's laws: $\neg(A \wedge B) = \neg A \vee \neg B$
 $\neg(A \vee B) = \neg A \wedge \neg B$

Step 3: Standardize Variables

Rename variables if necessary.

Step 4: Eliminate Quantifiers

(Used in FOL)

Step 5: Distribute OR over AND

Use distributive law.

Step 6: Obtain Clause Form

Represent expression as set of clauses.

Convert the Given Expression into CNF Expression: $(\neg P \vee Q) \rightarrow R$

Step 1: Remove Implication

Using: $A \rightarrow B \equiv \neg A \vee B$

We get: $\neg(\neg P \vee Q) \vee R$

Step 2: Apply DeMorgan's Law $(P \wedge \neg Q) \vee R$

Step 3: Distribute OR over AND $(P \vee R) \wedge (\neg Q \vee R)$

Final CNF Form - $(P \vee R) \wedge (\neg Q \vee R)$

Clause Form - Clauses are: 1. $(P \vee R)$, 2. $(\neg Q \vee R)$

Q6b) What are the reasoning systems for Categories? Explain Semantic Network and Description Logic.

Reasoning Systems for Categories help AI systems organize and infer knowledge about objects and classes.

They support: Classification, Inheritance, Relationship reasoning

Common reasoning systems are: Semantic Networks, Description Logic, Frames, Ontologies

1. Semantic Network - A Semantic Network is a graphical knowledge representation technique where: Nodes represent objects or concepts, Links represent relationships

Structure of Semantic Network - Animal

Types of Relations

Relation Meaning

IS-A Category relationship

HAS-A Property relationship

PART-OF Component relationship

|
IS-A
|
Bird
|
IS-A
|
Parrot

Example -

Bird ----can----> Fly

|
IS-A
|

Parrot Parrot inherits property "can fly".

Advantages of Semantic Networks - Easy visualization, Supports inheritance, Simple representation

Disadvantages - Ambiguity in large networks, Difficult formal reasoning

Applications - NLP, Expert systems, Knowledge representation

2. Description Logic (DL) - Description Logic is a formal knowledge representation language used to describe: Concepts, Roles, Individuals

It is foundation of: Semantic Web, OWL ontology language

Components of Description Logic -

Example - Student $\subseteq$ Person

Meaning: "All students are persons."

Features of Description Logic -

Component	Meaning
Concept	Class/category
Role	Relationship
Individual	Specific object

Formal semantics, Supports automated reasoning, Handles hierarchies and constraints

Advantages - Precise representation, Efficient reasoning, Used in ontology engineering

Applications - Semantic Web, Medical diagnosis systems, Intelligent information retrieval, Ontology development

Resolution is a powerful inference mechanism used in AI reasoning. CNF conversion is necessary for applying resolution. Semantic Networks and Description Logic are important reasoning systems used for representing categories and relationships in knowledge-based systems.

Q5a) Explain Forward Chaining and Backward Chaining with properties, advantages and disadvantages.

Forward Chaining is a data-driven inference technique in which reasoning starts from known facts and applies inference rules to derive new facts until the goal is reached.

Working of Forward Chaining

1. Start with known facts.
2. Apply matching rules.
3. Generate new facts.
4. Continue until conclusion is obtained.

Rule Format - IF condition
THEN action/conclusion

Example - Facts *Rain*

Rule *Rain* → *WetRoad*

Since *Rain* is true: *WetRoad* is inferred.

Algorithm of Forward Chaining

1. Add known facts into KB
2. Match rules whose premises are true
3. Fire rules
4. Add new facts to KB
5. Repeat until goal achieved

Properties of Forward Chaining - Data-driven approach, Bottom-up reasoning, Starts from available facts, Generates all possible conclusions

Advantages - Simple and automatic reasoning, Suitable when large amount of data is available, Finds all reachable conclusions

Disadvantages - May generate unnecessary facts, Slower for large rule sets, High memory usage

Applications - Expert systems, Monitoring systems, Medical diagnosis

Backward Chaining is a goal-driven inference technique where reasoning starts from the goal and works backward to find supporting facts.

Working of Backward Chaining

1. Start with goal.
2. Find rules that conclude goal.
3. Check whether premises are true.
4. Continue backward until facts are found.

Example - Goal $NeedsMedicine(Ram)$ **Rules** $Sick(x) \rightarrow NeedsMedicine(x)$ $Fever(x) \rightarrow Sick(x)$ **Fact** $Fever(Ram)$

Reasoning

From: $NeedsMedicine(Ram)$

Need: $Sick(Ram)$

Need: $Fever(Ram)$

Fact exists, hence goal proved.

Algorithm of Backward Chaining

1. Start with goal
2. Search rules producing goal
3. Check premises
4. Recursively prove premises
5. Stop when facts are found

Properties of Backward Chaining - Goal-driven approach, Top-down reasoning, Searches only relevant rules, Efficient for specific queries

Advantages - Efficient for goal-oriented problems, Reduces unnecessary inference, Requires less memory

Disadvantages - Not suitable for many possible goals, Recursive search may become complex, Depends heavily on goal specification

Forward chaining derives conclusions from available facts. Backward chaining starts from goal and searches supporting facts. Both are widely used inference techniques in AI systems.

Q5b) Explain:

i) Unification in FOL

Unification is the process of making two logical expressions identical by finding suitable substitutions for variables.

The substitution obtained is called: Unifier, Most General Unifier (MGU)

Need of Unification - In First Order Logic: Variables may differ. Resolution requires matching predicates. Unification helps in matching expressions.

Example -

Consider: $Knows(x, John)$ and $Knows(Ram, y)$

Substitutions: $x = Ram$ $y = John$

After substitution: $Knows(Ram, John)$

Expressions become identical.

Steps in Unification

1. Compare predicates
2. Compare arguments one by one.
3. If variable found: Substitute suitable term.
4. Continue until: Expressions become identical OR Conflict occurs.

Advantages - Simplifies reasoning, Supports automated theorem proving, Important for resolution in FOL

Applications - Logic programming, Expert systems, NLP, Prolog

ii) Reasoning with Default Information

Default Reasoning allows AI systems to make assumptions based on typical situations when complete information is unavailable.

It is also called: Non-monotonic reasoning

Example -

General rule: Birds can fly

Default assumption: $Bird(x) \rightarrow Fly(x)$

But exceptions exist: $Penguin(x) \rightarrow \neg Fly(x)$

Working - If system knows: Tweety is a bird then assume: Tweety flies unless contrary information is available.

Features of Default Reasoning - Uses assumptions, Handles incomplete information, Supports exceptions

Advantages - Mimics human reasoning, Handles uncertain knowledge, Useful in real-world AI systems

Disadvantages - May lead to incorrect assumptions, Difficult exception handling, Complex reasoning mechanisms

Applications - Expert systems, Medical diagnosis, NLP, Intelligent agents

Q5b) Explain Unification Algorithm in FOL. Solve stepwise with proper comments whether $p(x, g(x))$ is equal to or not equal to $p(f(y), f(a))$

Given Expressions - **Expression 1:** $p(x, g(x))$, **Expression 2:** $p(f(y), f(a))$

We check whether both expressions can unify.

Step 1: Compare Predicate Symbols

Both predicates are: p So predicate names match. Proceed further.

Step 2: Compare First Arguments

First arguments: x and $f(y)$

Possible substitution: $x = f(y)$

Store substitution.

Step 3: Compare Second Arguments

Second arguments: $g(x)$ and $f(a)$

Substitute: $x = f(y)$

Then: $g(f(y))$

must match: $f(a)$

Step 4: Compare Function Symbols

Function symbols are: g and f

Since: $g \neq f$ function symbols do not match.

Hence unification fails.

Hence $p(x, g(x))$ and $p(f(y), f(a))$ are **NOT UNIFIABLE**.

Because after substitution: $g(f(y)) \neq f(a)$.

Different function symbols prevent matching.

Q5b) Write and explain the steps of Knowledge Engineering Process.

Knowledge Engineering is the process of creating a knowledge-based system by acquiring, organizing, representing, and using knowledge for reasoning and problem solving.

It is mainly used in: Expert systems, Intelligent agents, AI applications

Steps of Knowledge Engineering Process

1. Identify the Task

Determine: Problem domain, Objectives, Requirements

Example: Medical diagnosis, game playing, robotics.

2. Gather Relevant Knowledge

Collect: Facts, Rules, Expert knowledge, Domain information

Sources: Human experts, Books, Databases

3. Choose the Vocabulary

Define: Predicates, Constants, Functions, Relations

Example - $Human(x), Mortal(x)$

4. Encode General Knowledge

Represent general rules in logical form.

Example - $\forall x(Human(x) \rightarrow Mortal(x))$

5. Encode Specific Facts

Store specific information about objects.

Example - $Human(Ram)$

6. Query the Knowledge Base

Ask questions to the system.

Example - $Mortal(Ram)?$

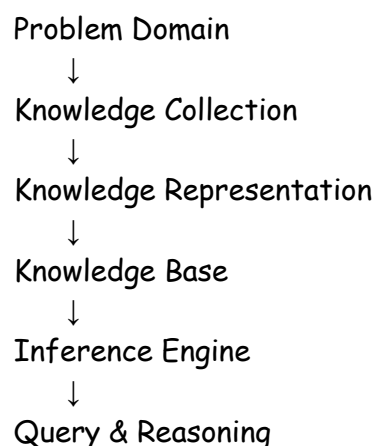
Inference engine derives answer.

7. Debug and Improve the Knowledge Base

Check: Correctness, Completeness, Consistency

Modify rules if required.

Diagram of Knowledge Engineering Process -



Advantages - Organized knowledge representation, Better reasoning capability, Reusable knowledge, Supports intelligent decision making

Applications - Expert systems, Robotics, NLP, Medical diagnosis

Q5b) What are the reasoning patterns in propositional logic? Explain them in detail.

Reasoning in propositional logic means deriving new conclusions from existing facts using logical rules.

It is used in: Expert systems, Automated theorem proving, AI reasoning systems

Common Reasoning Patterns in Propositional Logic

1. Modus Ponens

Rule - If: $P \rightarrow Q$ and P

Then infer: Q

Example - "If it rains, roads become wet."

$Rain \rightarrow WetRoad$
 $Rain$

Therefore: $WetRoad$

2. Modus Tollens

Rule - If: $P \rightarrow Q$ and $\neg Q$

Then infer: $\neg P$

Example - "If it rains, roads become wet."

Road is not wet: $\neg WetRoad$

Therefore: $\neg Rain$

3. AND Introduction

Rule - From: P and Q

Infer: $P \wedge Q$

Example - $Sunny$ Hot

Infer: $Sunny \wedge Hot$

4. AND Elimination

Rule - From: $P \wedge Q$

Infer: P or Q

Example - $Sunny \wedge Hot$

Infer: $Sunny$

5. OR Introduction

Rule - From: P

Infer: $P \vee Q$

Example - $Rain$

Infer: $Rain \vee Cold$

6. Resolution - Resolution is an important reasoning rule used in theorem proving.

Rule - Given: $P \vee Q$ and $\neg Q \vee R$

Infer: $P \vee R$

Example: $A \vee B \quad \neg B \vee C$

Infer: $A \vee C$

7. Hypothetical Syllogism

Rule - If: $P \rightarrow Q$ and $Q \rightarrow R$

Then infer: $P \rightarrow R$

Example - "If it rains, roads become wet." "If roads become wet, traffic increases."

Therefore: "If it rains, traffic increases."

Applications - Automated theorem proving, Expert systems, Logic programming, AI reasoning systems

Q6 c) What do you mean by Ontology of Situation Calculus?

Ontology of Situation Calculus - Situation Calculus is a mathematical logic language used in AI to represent and reason about dynamic worlds where actions change situations.

Ontology of Situation Calculus defines the basic objects used to represent actions and changes.

Main Components of Situation Calculus Ontology

1. Situation - A situation represents a state of the world at a particular time.

Example: S_0 Initial situation.

2. Actions - Actions are events that change situations.

Example: Move, Pick, OpenDoor

3. Fluents - Fluents are properties whose values change with situations.

Example: DoorOpen, At(Robot,Room1)

Representation - Situation + Action $\rightarrow$ New Situation

Example: $Result(OpenDoor, S_0) = S_1$

Meaning: Applying action OpenDoor in situation S_0 produces new situation S_1 .

Applications - Robotics, Planning systems, Intelligent agents, Automated reasoning systems